

Introduction to Machine Learning

Lab 5: Compressive Sensing

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1 Motivation

- Construct sparse signals based on predefined dictionary
- Implement Compressive Sensing and get some feelings about the power of random projection

2 Tasks

1. Implement a data generation function, which takes a dictionary and the number of nonzero coefficients as its input, and outputs the data matrix. **Please ensure that the mean value of the output data is zero.**
2. Implement a random projection function, which takes the data matrix and the dimension of the output as its input, and outputs 1) the random projection matrix, and 2) the random projection result. **The elements of the random projection matrix obey to normal distribution or Bernoulli distribution.**
3. Implement a data recovery function for compressive sensing, which takes the random projection result and the pre-defined dictionary matrix as its input.
4. Visualize the covariance matrix of the original data and that of the projected data, and observe their difference with respect to the dimension of the projection.